

lecture 2:-

Negation of Mathematical Operations.

$$\begin{aligned}\neg(=) &= \neq \\ \neg(>) &= \leq \\ \neg(\geq) &= < \\ \neg(<) &= \geq \\ \neg(\leq) &= >\end{aligned}$$

Q:- Find Negation -
 $2+3=5$

Let $p = 2+3=5$ $\neg p = 2+3 \neq 5$.

2- Logical Conjunction. $\wedge$

"and".

Ex 6
PS Find Conjunction of.

Today is Friday
It is raining.

Let $p =$ Today is Friday
" $q =$ It is raining.

$p \wedge q =$ Today is Friday and, It is raining.

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

3- Logical Disjunction. $\vee$

OR.

either p or q.

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Ex 6 Find disjunction of.

Today is Friday
It is raining.

Let $p =$ Today is Friday
" $q =$ It is raining.

$p \vee q =$ Today is Friday. Or, It is raining.

4- Implication:

✓ if p then $q \Rightarrow p \rightarrow q$.
if p, q .

p is sufficient for q .

$q \rightarrow p$ ✓ q whenever $p \Rightarrow p \rightarrow q$.
X.

p whenever $q = q \rightarrow p$.

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

P-6.

If I win election then I will lower taxes.
 p q

P-7. If $2+3=6$ then I am happy.
 p q

5- Biconditional

$\leftrightarrow$

$p \leftrightarrow q$

p iff q .

if p then q & Conversely.

p is necessary & sufficient for q .

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

Ex 8

She is taking a Math / IHL class. Look at.

Ex 10
P → q.

You can take a flight iff you buy a ticket.
P q.

Implication: $P \rightarrow Q$.

$r \rightarrow \neg s$.

Substitution.

$P \rightarrow Q \rightarrow (E) Q$.

Converse:

$q \rightarrow p$.

$t \rightarrow \neg z$.

HW.

Contrapositive.

$\neg q \rightarrow \neg p$.

$\neg t \rightarrow \neg (\neg z)$.

$s \rightarrow \neg r$.

$\neg t \rightarrow z$.

Inverse:

$\neg p \rightarrow \neg q$.

$\neg \neg z \rightarrow \neg t$
 $z \rightarrow \neg t$.

HW.

Ex 9 :- Write Contrapositive, Converse & Inverse of the following.

"the home team wins whenever it is raining".
q p.

$p \rightarrow q$.

if it is raining the home team wins.
p q

Converse: $q \rightarrow p$.

if the home team win then it is raining.

Contrapositive: $\neg q \rightarrow \neg p$.

$$\text{Inverse} := \neg p \rightarrow \neg q.$$

Sessional Questions.

"The Monarchs ~~does not~~ wins whenever it is raining"
Contrapositive

Find Implicator, Converse & Inverse? ($\forall x$).

Compound Proposition.

Tautology :- A Compound Proposition which is always true.

Contradiction :- " " " " " false.

Contingency :- " " " Sometime true & otherwise false.

p	$\neg p$
T	F
F	T

$p \wedge \neg p$
F
F

$p \vee \neg p$
T
T

Contradiction -

Tautology.

$p \wedge q$

$p \vee q$

$p \rightarrow q$

$p \leftrightarrow q$

$p \wedge q \vee r$

$p \wedge q \wedge r$

$p \rightarrow q \rightarrow r$

Precedence :-

()
~
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$\neg$
 $\rightarrow$
 $\leftrightarrow$

Ex 1 :- find TT $(p \vee \neg q) \rightarrow (p \wedge q)$.

P	q	$\neg q$	$p \vee \neg q$	$p \wedge q$	$(p \vee \neg q) \rightarrow (p \wedge q)$
T	T	F	T	T	T
T	F	T	T	F	F
F	T	F	F	F	T
F	F	T	T	F	T

logical Equivalence.
 Two equivalent if $p \leftrightarrow q$ is a tautology.

Ex 2 $\neg(p \vee q)$ $\neg p \wedge \neg q$.

P	q	$p \vee q$	$\neg(p \vee q)$	$\neg p$	$\neg q$	$\neg p \wedge \neg q$	$\neg(p \vee q) \leftrightarrow \neg p \wedge \neg q$
T	T	T	F	F	F	F	T
T	F	T	F	F	T	F	T
F	T	T	F	T	F	F	T
F	F	F	T	T	T	T	T

Propositional logic Applications.
 - System Constructing - Puzzles - Boolean Searching.

1. Question in Session 1.
 2. At Variables.
 Values.

Values Variables.

$P = \{T, F, N\}$

$3^2 = 9$

$P = \{T, R, N\}$
 $3^3 \quad 3^2 \quad 3^1 \quad 3^0$
 $\quad \quad \quad \delta \quad P \quad Q.$
 $\quad \quad \quad T \quad T$
 $\quad \quad \quad T \quad R$
 $\quad \quad \quad T \quad N.$
 $\quad \quad \quad R \quad T$
 $\quad \quad \quad R \quad R$
 $\quad \quad \quad R \quad N$
 $\quad \quad \quad N \quad T$
 $\quad \quad \quad N \quad R$
 $\quad \quad \quad N \quad N$

$$3^2 = 9.$$

P12
EX15 System Consistency (Next Lecture).